ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET first paper. Section A (30): multiple choice, true or false and matching · Section B (70): scientific notation, acceleration, force, work and power, the atom, component symbols, conductors and insulators, colour codes and parallel resistors, binary numbers, OR and EX-OR gates, analogue and digital signals, fire safety, first aid and workshop tools.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. A non-programmable calculator may be used. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 The SI unit of resistance is the …

A volt. B ampere. C watt. D ohm.

1.2 Which particle has no electric charge?

A Proton B Neutron C Electron D Ion

1.3 A material that allows current to flow easily is called a/an …

A conductor. B insulator.
C semiconductor. D dielectric.

1.4 A value of 2,2 kΩ is equal to …

A 22 Ω. B 220 Ω. C 2 200 Ω. D 22 000 Ω.

1.5 Which component stores electric charge?

A Resistor B Capacitor C Diode D Fuse

1.6 An ohmmeter measures …

A resistance. B current. C voltage. D power.

1.7 Which gate gives an output of 1 only when its two inputs are different?

A AND B NOR C NAND D EX-OR

1.8 The decimal number 9 written in binary is …

A 1000. B 1011. C 1001. D 0110.

1.9 Which ONE of the following is a semiconductor material?

A Silicon B Copper C Rubber D Glass

1.10 Which type of fire extinguisher may be used on an electrical fire?

A Water
B Carbon dioxide (CO_2)
C Foam
D Wet blanket soaked in water

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 Electrons flow from the negative terminal of a battery, through the circuit, to the positive terminal.

2.2 A fuse that keeps blowing should be replaced with a thicker wire.

2.3 The first two colour bands of a resistor give its two significant figures.

2.4 An oscilloscope shows how a voltage changes over time.

2.5 Glass is a good conductor of electricity.

2.6 A charged capacitor blocks direct current.

2.7 A digital signal has only two levels.

2.8 Water may be used to put out an electrical fire.

2.9 Resistors connected in series carry the same current.

2.10 A diode conducts current equally well in both directions.

(10)

[10]

QUESTION 3

Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–M) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Forward bias	A	rate of change of velocity
3.2	Semiconductor	B	melts to protect a circuit when the current is too high
3.3	Electron	C	number system that uses only 0 and 1
3.4	Fuse	D	material whose conductivity is between that of conductors and insulators
3.5	LDR	E	heats a joint so that solder can flow
3.6	Switch	F	current flows because the anode is more positive than the cathode
3.7	Binary	G	opens or closes a circuit
3.8	Acceleration	H	measures current and is connected in series
3.9	Ammeter	I	negatively charged particle that orbits the nucleus

3.10	Soldering iron	J	resistance falls as the light gets brighter
		K	stores energy in a magnetic field
		L	mass per unit volume
		M	changes AC into DC

(10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 Write 4 500 000 in scientific notation. (1)

4.2 Write $7{,}2 \times 10^{-3}$ in decimal notation. (1)

4.3 A mobile robot speeds up from rest to 2 m/s in 8 seconds. Calculate its acceleration. (4)

4.4 Calculate the force needed to give a 5 kg robot an acceleration of 2 m/s². (2)

4.5 A robot pushes a box with a force of 50 N over a distance of 3 m. Calculate the work done. (3)

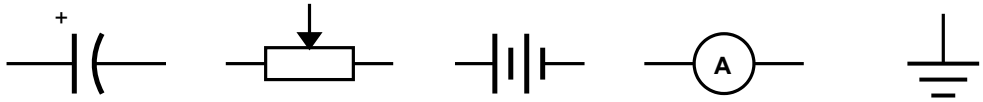
4.6 The work in QUESTION 4.5 is done in 5 seconds. Calculate the power. (3)

4.7 Name the THREE particles in an atom and give the charge of each. (6)

[20]

QUESTION 5

5.1 Identify each of the following component symbols:



5.1.1 5.1.2 5.1.3 5.1.4 5.1.5

(5)

5.2 Draw the symbol of a PNP transistor and label its THREE terminals. (5)

[10]

QUESTION 6

6.1 Explain the difference between a conductor and an insulator, and give ONE example of each. (4)

6.2 A resistor has the colour bands red, red, orange, silver. Write down its value and tolerance. (3)

6.3 A 100 Ω resistor and a 300 Ω resistor are connected in parallel. Calculate the total resistance. (3)

[10]

QUESTION 7

7.1 Convert the binary number 10110 to a decimal number. Show your working. (2)

7.2 Draw the IEC symbol and the truth table of each of the following two-input gates:

7.2.1 OR gate (5)

7.2.2 EX-OR gate (5)

7.3 Explain the difference between an analogue signal and a digital signal. (3)

[15]

QUESTION 8

8.1 Name THREE types of fire extinguishers, and state which ONE may be used on an electrical fire. (4)

8.2 Describe the first-aid steps to follow when a person has received an electric shock. (5)

8.3 Name THREE tools used in an electronics workshop, and give the use of each. (6)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER